

Modeling the detailed kinetics of mitochondrial cytochrome *c* oxidase: Catalytic mechanism and nitric oxide inhibition

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Pannala VR, Camara AK, Dash RK. Modeling the detailed kinetics of mitochondrial cytochrome *c* oxidase: Catalytic mechanism and nitric oxide inhibition. *J Appl Physiol* 121: 1196–1207, 2016. First published September 15, 2016; doi:10.1152/jappphysiol.00524.2016.—**Cytochrome *c* oxidase (CcO) catalyzes the exothermic reduction of O₂ to H₂O by using electrons from cytochrome *c*, and hence plays a crucial role in ATP production.** Although details on the enzyme structure and redox centers involved in O₂ reduction have been known, there still remains a considerable ambiguity on its mechanism of action, e.g., the number of sequential electrons donated to O₂ in each catalytic step, the sites of protonation and proton pumping, and nitric oxide (NO) inhibition mechanism. In this work, we developed a thermodynamically constrained mechanistic mathematical model for the catalytic action of CcO based on available kinetic data. The model considers a minimal number of redox centers on CcO and couples electron transfer and proton pumping driven by proton motive force (PMF), and accounts for the inhibitory effects of NO on the reaction kinetics. The model is able to fit well all the available kinetic data under diverse experimental conditions with a physiologically realistic unique parameter set. The model predictions show that: 1) the apparent *K_m* of O₂ varies considerably and increases from fully reduced to fully oxidized cytochrome *c* depending on pH and the energy state of mitochondria, and 2) the intermediate enzyme states depend on pH and cytochrome *c* redox fraction and play a central role in coupling mitochondrial respiration to PMF. The developed CcO model can easily be integrated into existing mitochondrial bioenergetics models to understand the role of the enzyme in controlling oxidative phosphorylation in normal and disease conditions.

proton pumping; oxygen reduction; half-saturation constant; kinetic modeling

NEW & NOTEWORTHY

*We developed a thermodynamically constrained mechanistic mathematical model for the kinetics of cytochrome *c* oxidase, incorporating several salient features, e.g., electron transfer, proton pumping, and nitric oxide inhibition. The model realistically and explicitly accounts for the thermodynamic dependence of the electron transport and proton translocation mechanisms through its rate constants. The model accurately describes the cytochrome *c* oxidase kinetics under both isolated mitochondria and purified enzyme conditions in the presence and absence of nitric oxide with a minimal number of kinetic parameters.*

THE CONSTANT SUPPLY OF ENERGY required for normal cellular function is generated in the form of ATP from substrate oxidation through mitochondrial electron transport chain (ETC) and oxidative phosphorylation (OxPhos). Cytochrome *c*

oxidase (CcO) is the terminal enzyme in the ETC that acts as an electron (e^-) sink and reduces molecular oxygen (O₂) to H₂O by a sequential one e^- oxidation of cytochrome *c*. The reduction of O₂ is a highly exergonic process, and it accompanies vectorial transport of protons (H⁺) across the inner mitochondrial membrane (IMM) to contribute to the generation of proton motive force (PMF) required for ATP synthesis. Being the terminal enzyme in the ETC (complexes I–IV), the CcO activity depends on the PMF generated by other ETC complexes. Moreover, it is also now well established that nitric oxide (NO) inhibits mitochondrial respiration by reversible binding with the heme proteins of CcO that has physiological and pathophysiological significance (21). Thus, CcO acts as a control point (rate-limiting enzyme) in OxPhos. Therefore, understanding the detailed mechanism of O₂ reduction by CcO and how NO inhibits its function is critical for understating mitochondrial energy metabolism in health and disease.

The dimeric mammalian CcO enzyme is controlled by both the nuclear and mitochondrial genomes (48). The mitochondrial genome encodes three subunits of the enzyme, which are mostly identical between different species and are essential for its activity (3). These three subunits contain four metal redox-active centers (Cu_A, Cu_B, hemes *a* and *a*₃) that participate in the catalytic cycle. Electrons entry in the enzyme from cytochrome *c* occurs via the diatomic Cu_A, which then rapidly equilibrates with heme *a* without any associated uptake or pumping of H⁺. Heme *a* lies near van der Waals' contact with the binuclear center formed by heme *a*₃ and Cu_B and sequentially donates one e^- to the binuclear center. O₂ binds at the binuclear center and is reduced to H₂O (11). In the process, CcO pumps four H⁺ across the IMM into the intermembrane space (IMS) for the generation of the PMF. Several mechanisms of H⁺ translocation by CcO have been proposed; however, a final conclusion has not been reached regarding the exact mechanism of H⁺ pumping (6, 39, 44). It was observed experimentally that the binuclear center of CcO also binds to NO both competitively and noncompetitively during its catalytic cycle with similar affinities as O₂; however, the details are not completely understood (1, 8).

The mechanism of O₂ reduction by CcO was studied extensively using different spectroscopic techniques to understand the reaction steps involved in the catalytic cycle. However, suitable kinetic data were not available on the enzyme catalytic mechanism, and, hence, none of the available CcO mathematical models (26, 54, 55) incorporate mechanistic details. Several modeling studies that describe mitochondrial OxPhos used a simple Michaelis-Menten type equation for CcO to describe its kinetics with or without giving any consideration to thermodynamics or associated H⁺ pumping mechanism (15, 25,

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30, 47, 57). CcO is regulated by both its reaction substrates (O_2 and cytochrome *c*) and products of ETC and OxPhos (PMF and ATP) (2). At increased PMF, which constitutes both the transmembrane electrical potential ($\Delta\Psi$) and proton gradient (ΔpH), or at increased ATP-to-ADP ratio, the CcO enzyme activity is diminished (17, 34, 52). Unfortunately, none of the developed CcO models include all these mechanistic details exclusively in their kinetic flux expressions. Although the mathematical model recently developed by Wilson and Vinogradov (55) considers some of these aspects, the model is not thermodynamically constrained based on reversible reaction formalism and lacks consistency in the H^+ pumping mechanism and missing a H^+ in the overall reaction kinetics (37). Specifically, the model used a phenomenological dependence on the lumped energy state *Q*, instead of the actual PMF, and did not account for any pH gradient across the IMM, thereby lacking the potential effects of matrix pH on the CcO kinetics (37). In addition, with the corrected CcO flux expression in the original model (55) and physiologically plausible parameter values, the model does not describe the available kinetic data, as previously reported (37). Similarly, a detailed model was developed by Krab et al. (26), which used 26 reactions to describe the CcO kinetics containing many unknown parameters. However, this model was not validated with any experimental data, thereby limiting its utility.

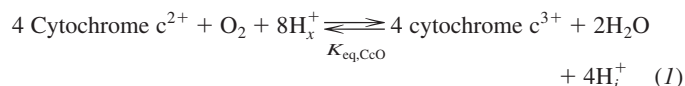
It has been recently observed experimentally that NO reversibly binds to the heme proteins of CcO and NO-dependent CcO inhibition is more potent under decreased O_2 tension (12). Under excessive NO generation, this would result in significant inhibition of CcO leading to depletion of ATP and contribute to pathology (43). Under physiological conditions, the NO-dependent inhibition of O_2 consumption at CcO around the endothelial source of NO results in increased O_2 levels, and allows its diffusion deeper in the tissue, thereby reducing hypoxic conditions (20). Thus, modeling the NO-mediated inhibition of CcO function is crucial to understand its role in both physiological and pathophysiological (e.g., hypoxia) conditions. However, none of the above-mentioned models include this inhibitory effect of NO on the CcO kinetics. Although Cooper and coworkers (14, 32) studied the detailed mechanisms of NO inhibition of mitochondrial respiration, they assumed a simple CcO kinetic model and did not account for H^+ pumping and thermodynamics, thus limiting its suitability in describing the CcO kinetics in mitochondria under different experimental conditions.

In the present study, we developed a detailed mathematical model for the CcO kinetics based on a hybrid catalytic scheme depending on several catalytic schemes presented in the literature (3, 22, 24, 49, 53–55). The model exclusively uses the midpoint potential information of each of the redox intermediates in the catalytic cycle that constitute the thermodynamic constraints. The model incorporates the details of the H^+ pumping mechanism for each e^- transfer step in the catalytic cycle and balances four H^+ involved in the CcO reaction. Furthermore, the model incorporates the details of reaction steps in the catalytic cycle where NO binds both competitively and noncompetitively to inhibit the CcO activity. The developed model effectively describes the available experimental data on the CcO kinetics with different cytochrome *c* reduced fraction (f_{red}) for varying $\Delta\Psi$ and pH (50) with a physiologically realistic unique parameter set. The model is also able to

qualitatively predict the dynamic profiles of O_2 concentration ($[O_2]$), O_2 consumption rate, and cytochrome *c* redox state for different $\Delta\Psi$ and pH conditions performed in a respirometer in the presence of artificial e^- donors (51). Furthermore, the model is able to describe the inhibitory effect of NO on the CcO activity for different values of f_{red} (32). The model predictions show that the apparent K_m of O_2 and V_{max} of CcO are functions of f_{red} , $\Delta\Psi$, and pH; apparent K_m of O_2 increases considerably with f_{red} depending on the values of $\Delta\Psi$ and pH. The intermediate enzyme states in the catalytic cycle are also functions of f_{red} , $\Delta\Psi$, and pH and play a central role in coupling mitochondrial respiration to the energy state. The model also shows CcO resides mainly in the O_2 -bound form under deenergized mitochondrial conditions, which is altered as mitochondria become more polarized leading to accumulation of other intermediate enzyme states, depending on NO concentration ($[NO]$). Therefore, the developed CcO model is mechanistic and thermodynamically constrained and suitable to be incorporated into other components of existing mitochondrial bioenergetics models (5, 35, 36, 38) to understand the role of the enzyme in controlling OxPhos in health and disease states, for example, ischemia-reperfusion injury.

METHODS

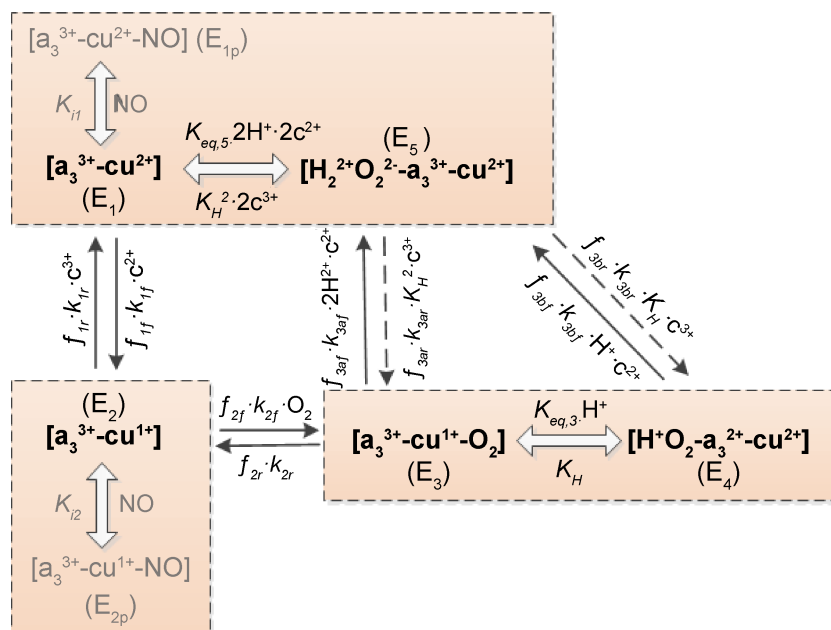
Hybrid catalytic scheme for the CcO reaction: CcO catalyzes the four e^- reduction of O_2 to H_2O using e^- from cytochrome *c* according to the following reaction:



where $K_{eq,CcO}$ is the apparent equilibrium constant for the CcO reaction. A total of eight positive charges are transferred across the IMM from matrix (x) to IMS (i). This includes four H^+ pumped from matrix to IMS, and four e^- transferred from IMS to matrix.

Our catalytic scheme for CcO is derived from Wilson and coworkers (53–55). Accordingly, the beginning steps of the catalytic pathway involving transfer of e^- from cytochrome *c* to Cu_A and the subsequent transfer to heme *a* to the binuclear center are assumed to be rapid and in thermodynamic equilibrium. Figure 1 shows a hybrid catalytic scheme for CcO showing the e^- transfer events starting from the binuclear center and the inhibitory interaction steps of NO on the enzyme. In the first step, cytochrome *c* donates one e^- through Cu_A and heme *a* to Cu_B in the oxidized binuclear center [$a_3^{3+}-Cu^{2+}$] (E_1) leading to one e^- reduced enzyme state [$a_3^{3+}-Cu^{1+}$] (E_2) with the coupled H^+ pumping from matrix to IMS. It is reported that as soon as one of the metal centers in the binuclear center is reduced, O_2 can enter the site and bind to the reduced metal atom (27), which is represented in Fig. 1 as [$a_3^{3+}-Cu^{1+}-O_2$] (E_3). At this stage, a rapid interchange of internal e^- distribution between Cu_B and heme a_3 takes place along with the binding of a H^+ leading to the enzyme state [$H^+O_2-a_3^{3+}-Cu^{2+}$] (E_4). Although states E_3 and E_4 are in rapid equilibrium with each other, interestingly, the next e^- in the sequence from cytochrome *c* can enter either state E_3 or E_4 depending on the O_2 binding position and number of H^+ on them. Because state E_4 is already in the protonated form, if e^- enters this state, it requires binding of one more H^+ to form the two H^+ -bound peroxide state [$H_2^+O_2^{2-}-a_3^{3+}-Cu^{2+}$] (E_5), whereas if it enters the state E_3 , it requires binding of two H^+ to terminate at the same two H^+ -bound peroxide state E_5 . Here, overall one e^- is donated by cytochrome *c* to state E_3 or E_4 leading to the state E_5 and is accompanied by one H^+ being pumped from matrix to IMS. In the final rapid equilibrium step, the tightly bound peroxide form of the enzyme (E_5) is further reduced by a two e^- transfer reaction along with the final two H^+ being pumped

Fig. 1. The proposed hybrid catalytic scheme for the reduction of O_2 by cytochrome *c* oxidase (CcO). In each catalytic cycle, cytochrome *c* donates one electron to the oxidized binuclear center (E_1) leading to the formation of one electron reduced binuclear center (E_2). O_2 binding takes place at the E_2 binuclear center, converting it into two rapid equilibrium intermediate enzyme states (E_3 and E_4) with internal electron distribution that is dependent on pH. The subsequent electron can enter either of the two states (E_3 and E_4) leading to the peroxide state (E_5). In the final rapid equilibrium step, two electron reduction of E_5 is accompanied by two protons uptake, and pumping generates the oxidized binuclear center (E_1) for the next cycle. Nitric oxide (NO) can bind both at the oxidized and reduced binuclear center, which are shown by E_{1p} and E_{2p} , respectively. Each orange box indicates the relevant rapid equilibrium states in the catalytic cycle.



from matrix to IMS. This step also includes the rapid equilibrium binding of two H^+ required for the formation of $2H_2O$ thereby leading to the initial oxidized binuclear center (E_1) (Fig. 1, $E_5 \rightarrow E_1$). Thus, for a full one cycle of CcO turnover, Fig. 1 shows the four e^- transfer steps from cytochrome *c* to O_2 with accompanied four H^+ being pumped from matrix to IMS and the binding of four H^+ used in the reduction of O_2 for the release of $2H_2O$ in the balanced CcO reaction shown in equation 1. To model the H^+ pumping mechanism, we assumed each e^- transfer reaction is accompanied by H^+ uptake from the matrix side and pumping across the IMM. The details of the H^+ pumping mechanism are incorporated in the apparent rate constants associated with each step shown in Fig. 1 and explained in the subsequent sections below. Furthermore, because NO can inhibit CcO by binding to both O_2 -bound and also to the oxidized form of the Cu_B in the binuclear center in competitive and uncompetitive modes (14, 32), we introduced these interactions in the CcO catalytic scheme as E_{2p} and E_{1p} , respectively. Because the model strictly follows mass conservation and thermodynamic laws, we assume all steps (reactions) in the catalytic scheme are reversible.

Kinetic flux expression for the CcO reaction. In Fig. 1, states shown in the orange boxes are in rapid equilibrium, and each box can be considered as a single state. At steady state, the rate of change of total enzyme concentration in each box is zero, and is solved along with the mass balance equation on the total enzyme concentration using the freely available KAPattern Matlab tool for generating rate equations (42) that result in the following flux expression for the CcO reaction (J_{CcO} , M/s):

$$J_{CcO} = \frac{E_t (k_{1f} f_{1f} k_{2f} f_{2f} k_{3f} C_r^2 C_{O_2} - k_{1r} f_{1r} k_{2r} f_{2r} k_{3r} C_o^2)}{k_{1f} f_{1f} k_{3f} C_o^2 + k_{1f} f_{1f} k_{3f} C_r^2 + k_{1f} f_{1f} k_{3f} C_o C_r + k_{1f} f_{1f} k_{2f} f_{2f} C_o + k_{2f} f_{2f} k_{3f} C_o + k_{1f} f_{1f} k_{2f} f_{2f} C_r + k_{2f} f_{2f} k_{3f} C_o C_{O_2} + k_{2f} f_{2f} k_{3f} C_r C_{O_2} + k_{1f} f_{1f} k_{2f} f_{2f} C_r C_{O_2}} \quad (2)$$

where E_t represents the total CcO concentration, C_r and C_o represent the reduced and oxidized cytochrome *c* concentrations, respectively, C_{O_2} represents the dissolved $[O_2]$, k_i represents the rate constants associated with each reaction, and f_i represents the fractional binding polynomials associated with each intermediate enzyme form in a particular rapid equilibrium state (see Fig. 1), which are defined as:

$$k_{3fp} = k_{3af} f_{3af} + k_{3bf} f_{3bf}; \quad k_{3rp} = (k_{3ar} + k_{3br}) f_{3r};$$

$$f_{1f} = \left(1 + \frac{1}{K_{eq,E5-1}} \left(\frac{C_o}{C_r} \right)^2 + \frac{C_{NO}}{K_{i1}} \right)^{-1}; \quad f_{1r} = f_{2f} = \left(1 + \frac{C_{NO}}{K_{i2}} \right)^{-1};$$

$$f_{2r} = f_{3af} = (1 + K_{eq,E3-4})^{-1}; \quad f_{3bf} = K_{eq,E3-4} (1 + K_{eq,E3-4})^{-1};$$

$$f_{3r} = \left(\frac{1}{K_{eq,E5-1}} \left(\frac{C_o}{C_r} \right)^2 + \frac{C_{NO}}{K_{i1}} \right) \left(1 + \frac{1}{K_{eq,E5-1}} \left(\frac{C_o}{C_r} \right)^2 + \frac{C_{NO}}{K_{i1}} \right)^{-1}; \quad (3)$$

where C_{NO} denotes $[NO]$, $K_{eq,i}$ indicates the apparent equilibrium constant for the associated rapid equilibrium reaction, and K_{i1} and K_{i2} are the dissociation constants for the NO binding at sites E_1 and E_2 , respectively. The CcO reaction flux J_{CcO} in equation 2 (in units of M/s) can be expressed as cytochrome *c* turnover number (TN) (in units of 1/s) by multiplying J_{CcO} by $4/C_t$, where C_t is the total cytochrome *c* concentration ($C_t = C_r + C_o$), and 4 is the number of e^- transferred to each molecule of O_2 .

Thermodynamic and kinetic constraints for the CcO reaction model parameters. The standard midpoint redox potentials for e^- transfer reactions and the corresponding equilibrium constants derived from the midpoint redox potentials (28, 29, 55) are described below:

$$E_{m,CuB}^0 = 0.35 \text{ V}; \quad E_{m,a3}^0 = 0.385 \text{ V}; \quad E_{m,cytc}^0 = 0.25 \text{ V};$$

$$E_{m,O_2}^0 = 0.85 \text{ V}; \quad K_{eq,E1-2}^0 = \exp \left(\frac{F(E_{m,CuB}^0 - E_{m,cytc}^0)}{RT} \right);$$

$$K_{eq,E3-4}^0 = K_3 \exp \left(\frac{F(E_{m,a3}^0 - E_{m,CuB}^0)}{RT} \right);$$

$$K_{eq,E4-5}^0 = \exp \left(\frac{F(E_{m,O_2}^0 - E_{m,cytc}^0 + E_{m,O_2}^0 - E_{m,a3}^0)}{RT} \right);$$

$$K_{eq,E3-5}^0 = K_3 \exp \left(\frac{F(E_{m,O_2}^0 - E_{m,cytc}^0 + E_{m,O_2}^0 - E_{m,CuB}^0)}{RT} \right);$$

$$K_{eq,E5-1}^0 = K_5 \exp \left(\frac{2F(E_{m,O_2}^0 - E_{m,cytc}^0)}{RT} \right);$$

$$K_{eq,CcO}^0 = \exp \left(\frac{4F(E_{m,O_2}^0 - E_{m,cytc}^0)}{RT} \right); \quad (4)$$

where E_m^0 represents midpoint redox potential at pH 7. Although slightly different values are reported for these midpoint redox potentials in the literature (33), we considered the values reported in Wilson and coworkers (28, 29, 55) to be consistent with their studies. F , R , and T represent Faradays constant (96,485 J·mol⁻¹·V⁻¹), ideal gas constant (8.314 J·mol⁻¹·K⁻¹), and temperature (298.15 K), respectively. K_3 and K_5 are the equilibrium constants associated with the respective internal e^- transfer reactions shown in Fig. 1. K_{eq}^0 indicates the equilibrium constant associated with each rapid equilibrium reaction.

The effects of H^+ binding and the associated H^+ pumping mechanisms are incorporated into the rate constants and equilibrium constants based on Eyring's free energy barrier formalism (56). These kinetic parameters along with the thermodynamic constraints are given by:

$$\begin{aligned} K_{eq,E1-2} &= K_{eq,E1-2}^0 e^{-\frac{FQ}{RT}}; & K_{eq,E2-3} &= \frac{k_{2f}}{k_{2r}}; \\ K_{eq,E3-4} &= K_{eq,E3-4}^0 \left(\frac{C_{Hx}}{K_H} \right); & K_{eq,E4-5} &= K_{eq,E4-5}^0 \left(\frac{C_{Hx}}{K_H} \right) e^{-\frac{FQ}{RT}}; \\ K_{eq,E3-5} &= K_{eq,E3-5}^0 \left(\frac{C_{Hx}}{K_H} \right)^2 e^{-\frac{FQ}{RT}}; \\ K_{eq,E5-1} &= K_{eq,E5-1}^0 \left(\frac{C_{Hx}}{K_H} \right)^2 e^{-\frac{2FQ}{RT}}; \\ K_{eq,CcO} &= K_{eq,CcO}^0 \left(\frac{C_{Hx}}{K_H} \right)^4 e^{-\frac{4FQ}{RT}}; & k_{1f} &= k_{1f0} e^{-\frac{\beta FQ}{RT}}; \\ k_{3af} &= k_{3af0} \left(\frac{C_{Hx}}{K_H} \right) e^{-\frac{\beta FQ}{RT}}; & k_{3bf} &= k_{3bf0} \left(\frac{C_{Hx}}{K_H} \right) e^{-\frac{\beta FQ}{RT}}; \\ k_{1r} &= \frac{k_{1f}}{K_{eq,E1-2}}; & k_{3ar} &= \frac{k_{3af}}{K_{eq,E3-5}}; & k_{3br} &= \frac{k_{3bf}}{K_{eq,E4-5}}; \\ K_5 &= \frac{K_{eq,CcO}}{K_{eq,E1-2} K_{eq,E2-3} K_{eq,E3-5} K_{eq,E5-1}}; \end{aligned} \quad (5)$$

where C_{Hx} is the matrix H^+ concentration, K_H is the H^+ binding dissociation constant, β is the height of the free energy barrier, and Q is the lumped energy state of mitochondria, which is correlated with the PMF based on the dependence of the CcO function on the ATP synthase activity as discussed below.

Relationship between the mitochondrial energy state Q and PMF. During OxPhos, one molecule of O_2 , two molecules of NADH, five molecules of ADP, and five molecules of inorganic phosphate (P_i) are consumed to synthesize five molecules of ATP, assuming a 2.5 P-to- O ratio for complex I-based substrates (e.g., pyruvate). In the process, 20 H^+ are pumped out of the matrix (8 at complex I, 8 at complex III, and 4 at complex IV), whereas 15 H^+ are pumped back into the matrix via F_1F_0 -ATPase (ATP synthase), and 5 H^+ are cotransported with 5 molecules of P_i into the matrix via $H_2PO_4^-$ - H^+ carrier (an electroneutral symporter), with concomitant 5 molecules of ADP^{3-} influx for ATP^{4-} efflux via adenine nucleotide translocase (ANT). In terms of the net positive charges transferred in the ETC across the IMM, eight net positive charges are transferred at complex I, but only four net positive charges are transferred at complex III because of the simultaneous transfer of four e^- from matrix to IMS via ubiquinol to cytochrome c. Similarly, at complex IV, four H^+ pumped out are accompanied by four e^- transfer from IMS to matrix via cytochrome c to O_2 . Therefore, the net positive charges transferred across the IMM by complex IV are eight. Because 20 positive charges transferred from the matrix to the IMS are associated with the synthesis of five molecules of ATP, eight net positive charges transfer at complex IV would account for two molecules of ATP synthesis. Therefore, each e^- transfer at complex IV (CcO) would correspond to 0.5 molecules of ATP produced. Assuming pumping of three H^+ via ATP

synthase would produce one molecule of ATP, the lumped energy state Q in equation 5 correlates with 1.5 times PMF:

$$Q = 1.5(\Delta\Psi + 2.303(RT/F)\Delta pH), \quad (6)$$

where $\Delta\Psi$ and ΔpH are membrane potential and pH gradient across the IMM, respectively.

Analytical expressions for the apparent K_m of O_2 and V_{max} of the CcO reaction. The analytical expressions for the apparent K_m of O_2 and V_{max} of the CcO reaction can be easily obtained by equating equation 2 with an equation of the form $J_{CcO} = V_{max}C_{O_2}/(K_m + C_{O_2})$. This way the K_m of O_2 and V_{max} are functions of the cytochrome c reduced fraction (f_{red}), mitochondrial energy state, and pH. Because the reverse flux of the CcO reaction is negligible (i.e., since $k_{1r}f_{1r}k_{2r}f_{2r}k_{3rp} \approx 0$), we have:

$$V_{max} = \frac{E_r k_{1f} f_{1r} k_{2f} f_{2r} k_{3fp} C_r^2}{k_{2f} f_{2r} k_{3rp} C_o + k_{2f} f_{2r} k_{3fp} C_r + k_{1f} f_{1r} k_{2r} C_r} \quad (7)$$

$$K_m = \frac{k_{1r} f_{1r} k_{3rp} C_o^2 + k_{1f} f_{1r} k_{3fp} C_r^2 + k_{1r} f_{1r} k_{3fp} C_o C_r + k_{1r} f_{1r} k_{2r} f_{2r} C_o + k_{2f} f_{2r} k_{3rp} C_o + k_{1f} f_{1r} k_{2r} f_{2r} C_r}{k_{2f} f_{2r} k_{3rp} C_o + k_{2f} f_{2r} k_{3fp} C_r + k_{1f} f_{1r} k_{2r} f_{2r} C_r} \quad (8)$$

Alternatively, the V_{max} can be obtained numerically by evaluating equation 2 at very high C_{O_2} , and the K_m of O_2 can be obtained numerically by solving the value of C_{O_2} at which the value of J_{CcO} is $V_{max}/2$ in equation 2.

Numerical estimation of the CcO reaction model parameters. The developed CcO kinetic model contains 14 unknown parameters (shown in Fig. 1), including the height of the free energy barrier (β). We used the values of the midpoint redox potentials and the corresponding equilibrium constants given in equation 4 to constrain the reverse reaction rate constants k_{1r} , k_{3ar} , and k_{3br} , as shown in equation 5. Similarly, we used the overall equilibrium constant of the CcO reaction to constrain the parameter K_5 , as shown in equation 5. Furthermore, we fixed the rate constants associated with the binding of O_2 (k_{2f}) and dissociation (k_{2r}) based on values found in the literature (16, 19, 53). We also fixed the H^+ binding constant K_H as 10^{-7} M based on the reference values of pH at which midpoint redox potentials are fixed. To estimate the remaining seven unknown kinetic parameters, we used a two-step optimization technique of the combined least-square objective function defined below to fit the model simulated outputs to the experimental data both in the absence (50) and presence (32) of NO:

$$\min_{\phi} E(\phi), \quad E(\phi) = \sum_{k=1}^{N_{exp}} \frac{1}{N_{data,k}} \left[\sum_{j=1}^{N_{data,k}} \left(\frac{J_{j,k}^{data} - J_{j,k}^{model}(\phi)}{\max(J_{j,k}^{data})} \right)^2 \right], \quad (9)$$

where N_{exp} is the number of experiments and $N_{data,k}$ is the number of data points in k th experiment; $J_{j,k}^{data}$ are the experimental data, and $J_{j,k}^{model}$ are the corresponding model simulations that depend on the model parameter values ϕ . The accuracy and robustness of the model fitting to the experimental data are assessed based on the value of the mean square residual error $E(\phi)$ and the sensitivities of $E(\phi)$ or $J(\phi)$ to perturbations in the optimal parameter estimates, respectively.

RESULTS

As discussed above, the unknown model parameters were estimated based on available experimental data both in the absence and presence of NO (32, 50). Wilson et al. (50) performed experiments in the absence of NO using isolated rat liver mitochondria and measured the CcO activity under diverse experimental conditions using artificial electron donors that directly reduce cytochrome c. They used N,N,N',N' -tetramethyl- p -phenylenediamine (TMPD) coupled with ascorbate as the electron donors to get different reduced

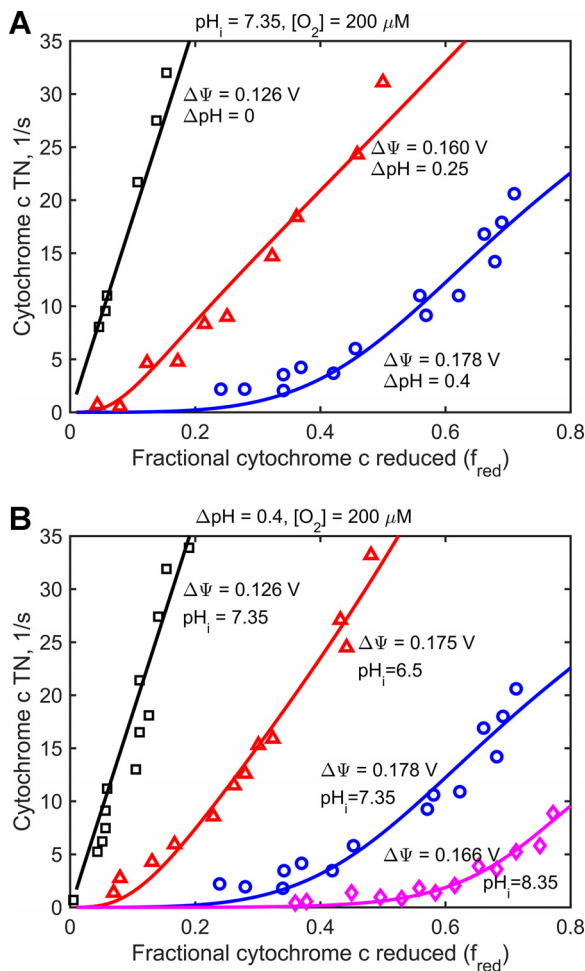


Fig. 2. Model simulations of the experimental data on the CcO kinetics. A: dependence of cytochrome *c* turnover number (TN) on the fraction of cytochrome *c* reduced (f_{red}) at pH 7.35 for three different levels of proton motive force (PMF): *p*-trifluoromethyl phenylhydrazone carbonyl cyanide (FCCP)-treated uncoupled (squares, black), maintained at the state 3 [ATP]/([ADP][P_i]) (where brackets denote concentration) level (triangle, red), and state 2 coupled (circles, blue). B: dependence of cytochrome *c* TN on the level of f_{red} under coupled conditions for three different levels of medium pH of 6.5 (triangles, red), 7.35 (circles, blue), and 8.35 (diamond, pink). Black squares with solid line represent experimental and model simulations for uncoupled conditions at pH 7.35. The concentrations of total cytochrome *c* and CcO used were 2 and 1 μ M, respectively, under saturating high P_{O_2} levels in the absence of NO. Symbols are experimental data (50), and the solid lines represent model simulations.

cytochrome *c* fractions (f_{red}) and measured O_2 consumption rate at different buffer pH. They also performed similar experiments with or without added ATP and *p*-trifluoromethyl phenylhydrazone carbonyl cyanide (FCCP) to identify the CcO function under different mitochondrial energy state (PMF or Q). In another study, Mason et al. (32) performed experiments in the presence of NO using purified beef heart mitochondrial CcO, and the reduction of O_2 by CcO was monitored for different turnover numbers (TN) of the CcO enzyme using TMPD and ascorbate coupled to get different levels of f_{red} . Fractional enzyme activities and apparent IC_{50} for NO were calculated for different [NO] and $[O_2]$, respectively.

Dependence of the CcO function on f_{red} , PMF, pH, and NO. Figure 2 shows the comparison of model simulations (solid

lines) with the experimental data (symbols) of Wilson et al. (50) in the absence of NO based on the estimated values of the model parameters that are presented in Table 1. Specifically, Fig. 2A shows the model fits to the data on cytochrome *c* TN as a function of f_{red} for different values of $\Delta\Psi$ (PMF or Q) at pH of 7.35 and for high P_{O_2} . The model was able to fit these data sets very well for isolated mitochondria both under uncoupled and coupled (states 2 and 3) conditions. We estimated the $\Delta\Psi$ values shown in Fig. 2 by assuming the ΔpH values 0, 0.25, and 0.4 for deenergized (0.126 V), ATP addition (0.16 V; state 3), and coupled (0.178 V; state 2) conditions, respectively. Both experimental and model simulation results clearly show that the CcO flux (cytochrome *c* TN) is maximum under uncoupled conditions and requires very low levels of f_{red} (black squares and corresponding solid lines in Fig. 2, A and B). On the other hand, under coupled conditions with or without added ATP, the enzyme TN was considerably decreased even at higher values of f_{red} , indicating the effect of proton back pressure (high PMF) on the activity of CcO. It was also shown experimentally that the CcO activity varies significantly with the medium pH (50). As shown in Fig. 2B, the model was able to fit the experimental observations very well for three different pH values (6.5, 7.35, and 8.35) under well-coupled mitochondrial conditions ($\Delta\Psi \approx 0.175$ V; $\Delta pH \approx 0.25$). It can be seen from Fig. 2 that, under acidic conditions, CcO functions better compared with the alkaline conditions where cytochrome *c* TN is very small even for the higher levels of f_{red} .

It is well known that NO inhibits the CcO activity under nanomolar [NO] (32). The CcO model based on the catalytic scheme shown in Fig. 1 was able to fit the experimental data very well by estimating only two unknown parameters (K_{i1} and K_{i2}) related to the NO inhibition and keeping the rest of the parameters fixed as shown in Table 1. Figure 3A shows the fractional CcO activity as a function of [NO] for different values of CcO TN (f_{red}). The model simulations (solid lines) describe very well the experimental data (symbols) for both high and low CcO TN (f_{red}) and strengthens the argument that the inhibitory action of NO on the CcO activity depends on the CcO TN (f_{red}). NO strongly inhibits the CcO activity when cytochrome *c* is highly reduced. A similar result was obtained when apparent IC_{50} of NO inhibition was measured with varying levels of O_2 for different cytochrome *c* TN. The model was able to accurately fit the experimental data (Fig. 3B). The results clearly show that the NO inhibition of the CcO activity is highly dependent on the level of cytochrome *c* reduction (f_{red}).

Table 1. Parameter values used for the CcO kinetic model

Parameter	Value	Source
k_{10} , $M^{-1} \cdot s^{-1}$	8×10^9	Estimated
k_{2f} , $M^{-1} \cdot s^{-1}$	6×10^8	Ref. 16
k_{2r} , s^{-1}	10	Ref. 55
K_3	17.5	Estimated
k_{3af0} , $M^{-1} \cdot s^{-1}$	1.3×10^{10}	Estimated
k_{3bf0} , $M^{-1} \cdot s^{-1}$	6.1×10^8	Estimated
K_{i1} , M	40×10^{-9}	Estimated
K_{i2} , M	7×10^{-9}	Estimated
K_H , M	1×10^{-7}	Fixed
β	0.3	Estimated

CcO, cytochrome *c* oxidase; *k*, constant; β , height of the free energy barrier.

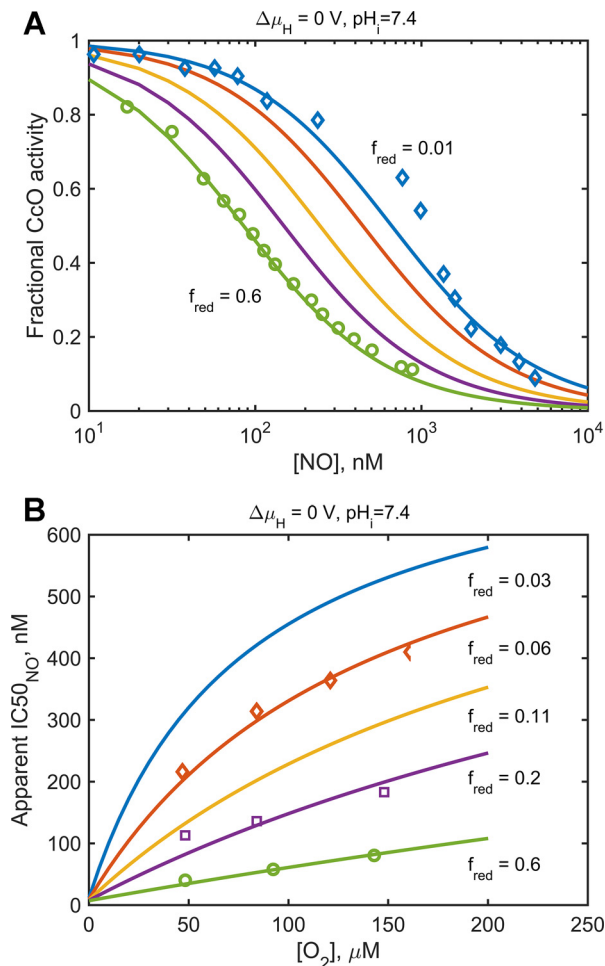


Fig. 3. Model simulations of experimental data on the inhibitory effects of NO on CcO kinetics. A: fractional CcO activity as a function of [NO] with varying cytochrome *c* reduction fraction f_{red} (or cytochrome *c* TN) from a low value 0.01 (blue) to a high value 0.6 (green) at pH 7.4 using purified CcO measured at a high saturating P_{O_2} . Red (0.05), yellow (0.15), and purple (0.3) lines represent the model results for the intermediate f_{red} values for which no experimental data were available. B: apparent IC₅₀ for NO as a function of [O₂] at different levels of f_{red} at pH 7.4 using purified CcO. The concentrations of total cytochrome *c* and CcO used were 1 and 0.5 μM, respectively. Symbols indicate experimental data (32), and solid lines represent model simulations.

The parameterized CcO model (in the absence of NO) was used to simulate the rate of O₂ consumption ($\dot{V}_{O_2}$) for several different experimental scenarios (Fig. 4, A–C). These simulations provide insights into the values of apparent K_m of O₂ and V_{max} of CcO under diverse experimental conditions. Specifically, the model was used to simulate the steady-state $\dot{V}_{O_2}$ as a function of P_{O_2} for different levels of f_{red} at fixed PMF of 0.17 V and pH of 7.35 (Fig. 4A), for different levels of PMF at fixed f_{red} of 0.2 and pH of 7.35 (Fig. 4B), and for different levels of pH at fixed PMF of 0.17 V and f_{red} of 0.2 (Fig. 4C). As expected, $\dot{V}_{O_2}$ was higher at higher f_{red} , which decreased as f_{red} decreased. In contrast, $\dot{V}_{O_2}$ was higher at lower PMF or pH, which decreased as PMF or pH increased. Under in vivo like mitochondrial conditions (high PMF and low f_{red}), $\dot{V}_{O_2}$ is significantly lower with very high apparent K_m of O₂ for CcO.

The developed CcO model is further used to ascertain the kinetics of the CcO enzyme at different pH by simulating the dynamic changes of P_{O_2} , $\dot{V}_{O_2}$, and f_{red} in a closed respirometer

under fixed PMF condition, as shown experimentally by Wilson et al. (51) (Fig. 4, D–F). For these simulations, to regenerate the reduced cytochrome *c* due to the CcO reaction, we used the artificial electron donor TMPD in combination with ascorbate to provide a continuous supply of electrons to the oxidized cytochrome *c*. The following set of differential equations was used to simulate the dynamic profiles:

$$\begin{aligned} \frac{dC_r}{dt} &= -4J_{CcO} + k_{rT}[TMPD]C_o \\ \frac{dC_{O_2}}{dt} &= -J_{CcO} \end{aligned} \quad (10)$$

where k_{rT} ($4.2 \times 10^4 \text{ M}^{-1} \cdot \text{s}^{-1}$) is the second-order rate constant for TMPD binding to oxidized cytochrome *c* (C_o) to regenerate C_r (31) and [TMPD] is TMPD concentration. These simulations clearly show how the CcO enzyme functions under dynamically changing mitochondrial conditions. The simulations show that, at a given initial high P_{O_2} in the chamber, the initial pseudo-steady-state level of f_{red} [at time (t) = 0 s] is dependent on the level of pH, and the steady-state level of f_{red} is increased with increased pH (Fig. 4, D–F). However, over time f_{red} slowly increased as the available O₂ is decreased, and the cytochrome *c* is suddenly completely reduced ($f_{red} = 1$) as O₂ is completely depleted from the chamber. The initial pseudo-steady-state level of f_{red} is higher and also the time of complete depletion of O₂ from the chamber is higher for higher (alkaline) pH (Fig. 4, E and F). These results indicate that, under alkaline conditions, the CcO activity decreases in accordance with Fig. 4C, which is shown by the progressive slow drops in P_{O_2} and $\dot{V}_{O_2}$ over time as seen in Fig. 4, E and F.

Apparent V_{max} and K_m of O₂ for CcO. The simulations in Fig. 4 clearly indicate that both apparent V_{max} and K_m of O₂ for CcO are significantly changed with varying f_{red} , PMF, and pH in the absence of NO. To understand the effects of these key variables for the entire range, in the presence and absence of NO, we calculated the apparent V_{max} and K_m of O₂ for CcO from the model using equations 7 and 8 (Fig. 5). The model simulations show that the apparent V_{max} is maximal at low pH (6.5) in the absence of NO (Fig. 5A, green) and is decreased as pH is increased to alkaline values [7.35 (red) and 8.35 (blue)]. In the presence of NO (Fig. 5C), the apparent V_{max} is further decreased at fixed pH (7.35) with increasing [NO] [100 (blue) and 500 (green) nM]. Interestingly, both in the presence and absence of NO, the apparent V_{max} is a function of f_{red} and PMF, which drastically decreased under low f_{red} and high PMF conditions. Similarly, the model simulations show that, in the absence of NO, the apparent K_m of O₂ is considerably higher at low f_{red} and high PMF, which decreased as PMF is decreased (Fig. 5B). Furthermore, at pH 6.7 (Fig. 5B, green), the apparent K_m of O₂ is lower for the combination of low f_{red} and high PMF compared with the alkaline pH values of 7.35 (red) and 8.35 (blue), but the trend was altered as PMF is decreased. In the presence of NO, at a constant pH of 7.35, the model simulations show that the apparent K_m of O₂ is considerably higher at low f_{red} and high PMF. However, the values were increased with increased [NO] [0 (red), 100 (blue), and 500 (green) nM] when PMF was decreased, indicating a crucial role of NO in the kinetics of CcO.

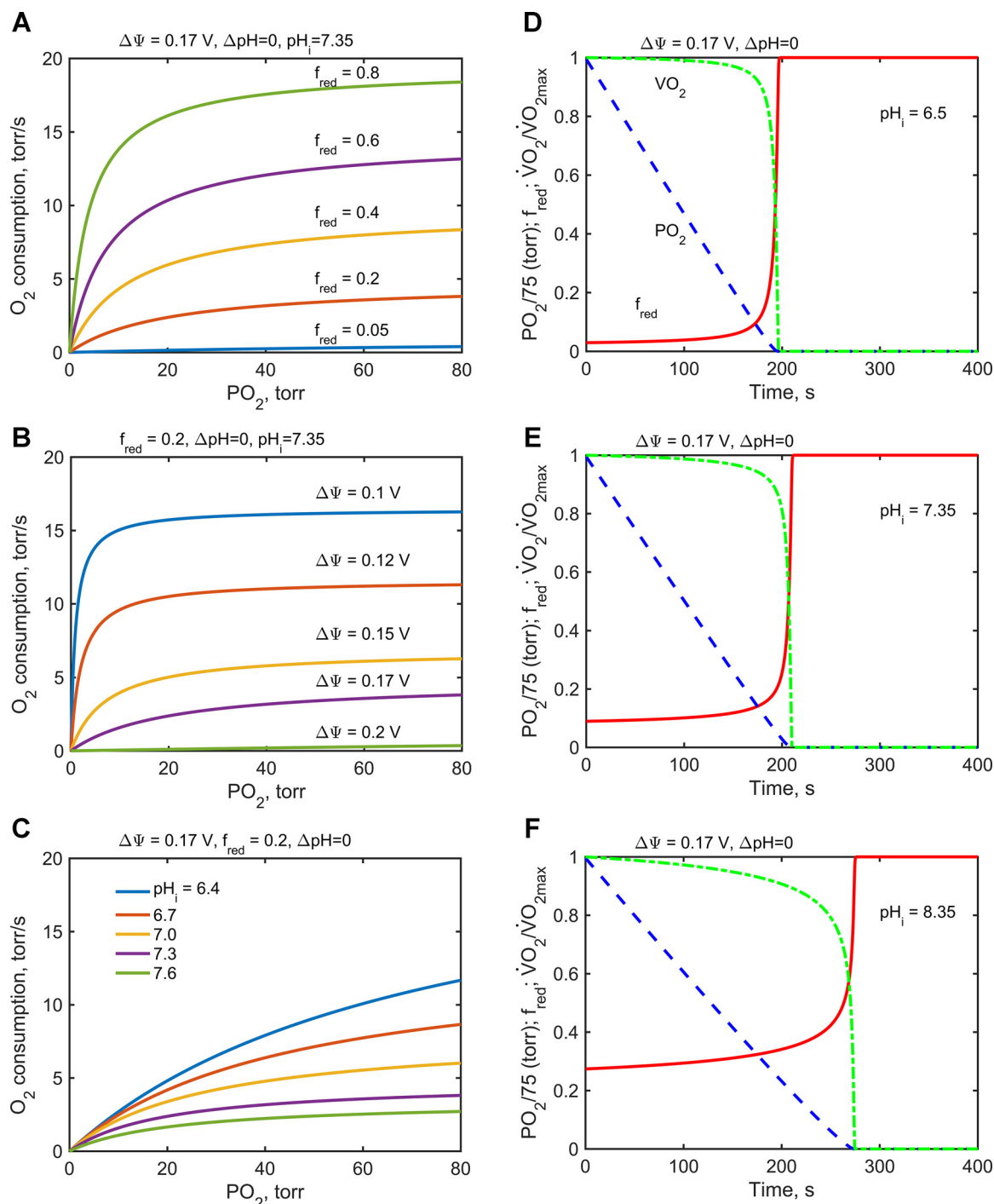


Fig. 4. CcO kinetic model predictions under diverse experimental conditions. A–C: model predictions of the initial pseudo-steady-state rate of O_2 consumption ($\dot{V}O_2$) as a function of PO_2 for five different levels of f_{red} of 0.05, 0.2, 0.4, 0.6, and 0.8 at a constant PMF of 0.17 V and medium pH of 7.35 (A), for five different levels of PMF of 0.1, 0.12, 0.15, 0.17, and 0.2 V at a constant f_{red} of 0.2 and medium pH of 7.35 (B), and for five different levels of medium pH of 6.4, 6.7, 7.0, 7.3, and 7.6 at a constant PMF of 0.17 V and f_{red} of 0.2 (C). All simulations were performed with total cytochrome *c* and CcO concentrations of 2 and 1 μ M, respectively. D–F: model predictions of the dynamics of PO_2 (blue), $\dot{V}O_2$ (green), and cytochrome *c* f_{red} (red) with CcO reaction in a closed respirometer supplemented with artificial electron donors *N,N,N',N'*-tetramethyl-*p*-phenylenediamine (TMPD, 70 μ M) and ascorbate under constant PMF and different medium pH of 6.5, 7.35, and 8.35. All model simulations were performed with total cytochrome *c* and CcO concentrations of 1 and 0.5 μ M, respectively, which corresponds to the experimental conditions of Wilson et al. (51).

Role of intermediate CcO states in determining the rate of CcO reaction. Given the dependence of CcO function on f_{red} , PMF, pH, and [NO], it was interesting to explore how the individual enzyme states in the catalytic cycle change with

respect to all of these variables (Fig. 6). In the absence of NO (Fig. 6, left), the model simulations for low pH (6.7) show that the enzyme largely resides in the intermediate state E_5 (blue) for the combination of low f_{red} and high PMF values but

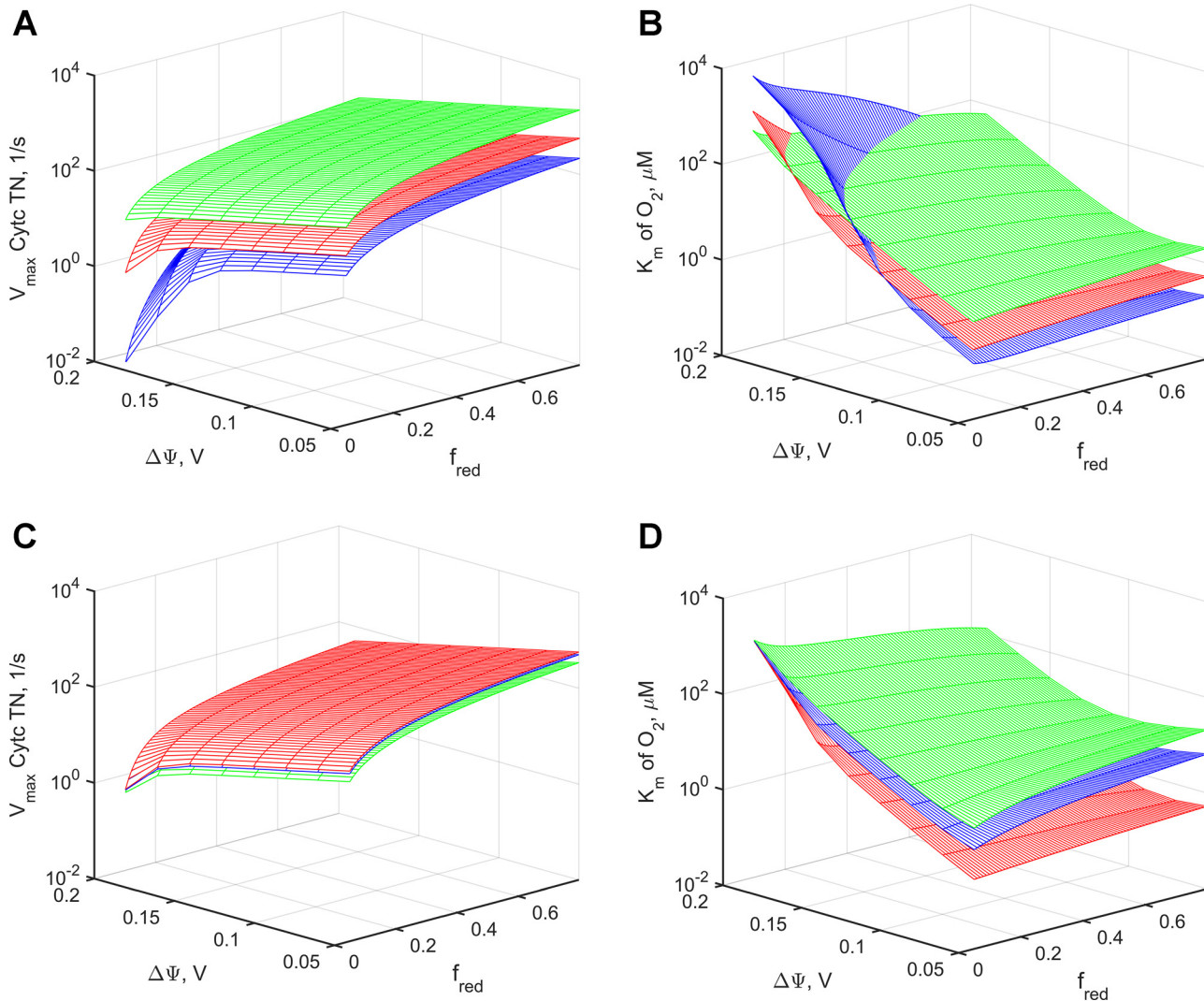


Fig. 5. The CcO kinetic model predictions of the apparent $V_{\max}$ and K_m of O_2 in cytochrome *c* TN in the absence and presence of NO. The 3-dimensional surface plots show the model predictions of the apparent $V_{\max}$ and K_m of O_2 for CcO as a function of $\Delta\Psi$ (or PMF with $\Delta pH = 0$) and cytochrome *c* reduction fraction (f_{red}) for three different levels of pH of 6.5 (green), 7.35 (red), and 8.35 (blue) in the absence of NO (A and B) and for three different [NO] of 0 (red), 100 (blue), and 500 (green) nM for a constant level of pH of 7.35 (C and D). All model simulations were performed with total cytochrome *c* and CcO concentrations of 2 and 1 μM , respectively.

switches to other states as PMF is decreased below 0.2 volts (Fig. 6A). In contrast, with decreasing values of PMF, the intermediate state E_4 (green) continuously increases, reaching its maximum value under low PMF irrespective of the f_{red} levels, indicating its role in determining the rate of O_2 reduction. A distinct behavior is observed for the oxidized binuclear center, E_1 (red); the enzyme state reaches its maximal value between PMF of 0.15 and 0.2 volts and then decreases when PMF is decreased below 0.15 volts. For higher pH values (Fig. 6, B and C), although the overall behavior of the intermediate enzyme states looks similar, the maximum E_1 (red) level is decreased as pH is increased from 6.5 to 8.35, whereas the intermediate state E_3 (yellow) is slightly built up at high pH (Fig. 6C), indicating the equilibrium shift between E_3 and E_4 . Interestingly, in the presence of 100 nM of NO (Fig. 6, right), these behaviors are completely altered, although the intermediate state E_5 is still maximum for the combination of low f_{red} and high PMF values, but it does not decrease to zero as PMF

is decreased to a low value (Fig. 6D). Rather, the NO-bound intermediate state E_{1p} (cyan) is increased as PMF is decreased below 0.2 volts, indicating the decrease in the amount of active enzyme available for the cytochrome *c* TN. In the case of alkaline pH (Fig. 6, E and F), the intermediate state E_1 is further decreased with decreasing PMF values, leading to the increased NO-bound states E_{1p} and E_{2p} (black) and increased intermediate state E_3 . This results in lowering the overall amount of active enzyme available for the cytochrome *c* TN, thereby leading to the observed low cytochrome *c* TN under alkaline conditions.

DISCUSSION

Mitochondrial OxPhos is a tightly controlled process based on the metabolic energy supply and demand. To meet high energy demands under various stress conditions, mitochondria need to have an adjustable control mechanism with

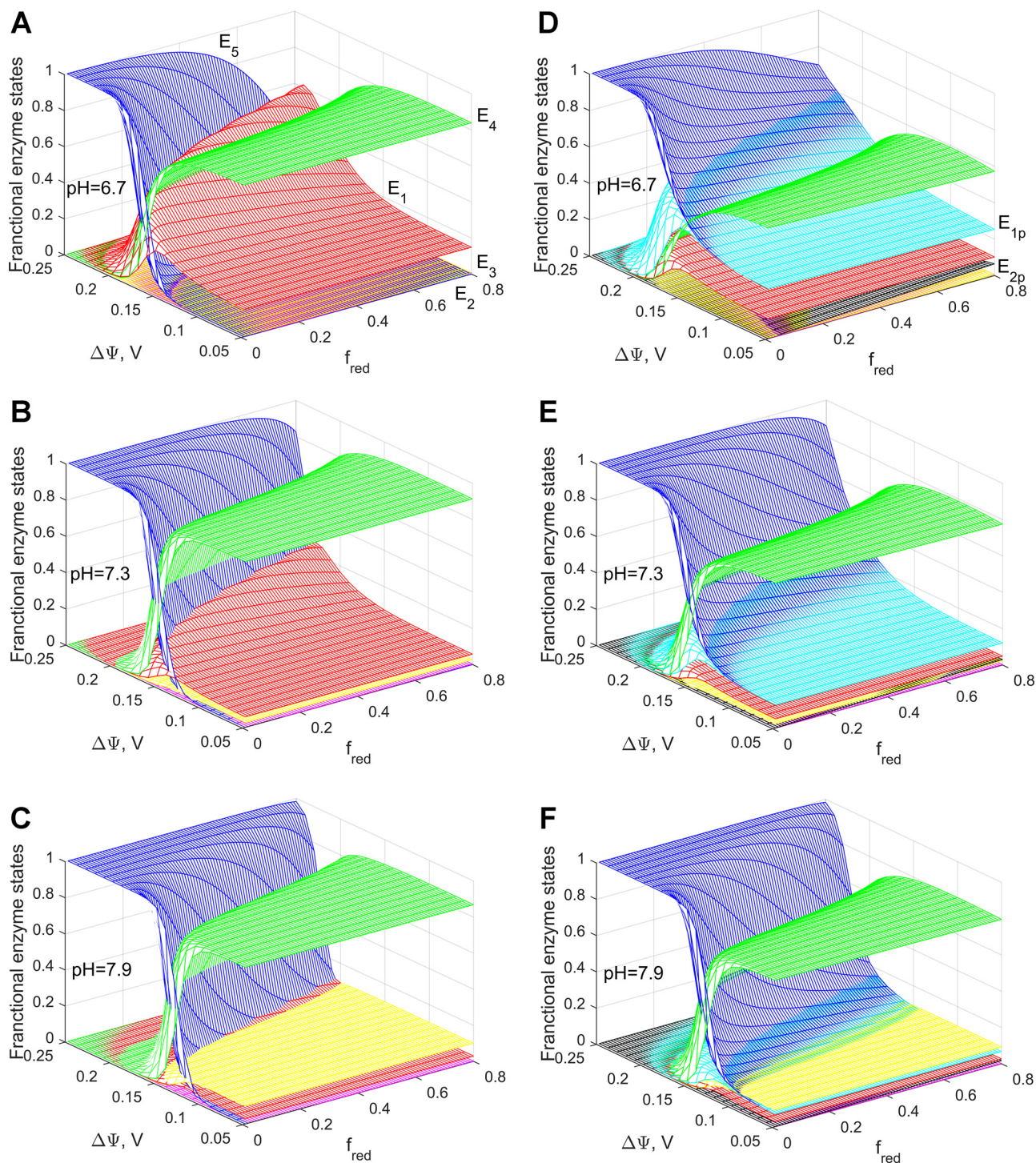


Fig. 6. CcO kinetic model predictions of the intermediate enzyme states in cytochrome *c* TN in the absence and presence of NO. The 3-dimensional surface plots show the model predictions of the fractional intermediate CcO states as a function of $\Delta\Psi$ (or PMF with $\Delta pH = 0$) and cytochrome *c* f_{red} for three different levels of pH of 6.7 (A and D), 7.3 (B and E), and 7.9 (C and F) in the absence (A–C) and 100 nM presence (D–F) of NO. In A–F, the enzyme states plotted are E₅ (blue), E₄ (green), E₃ (yellow), E₂ (magenta), E_{2p} (black), E₁ (red), and E_{1p} (cyan). All model simulations were performed with total cytochrome *c* and CcO concentrations of 2 and 1 μ M, respectively.

variable capacity for ATP synthesis. CcO, the terminal electron acceptor in the ETC, which is well explored spectrophotometrically, has the variable capacity to reduce O₂ depending on the demand for ATP. Several kinetic schemes were proposed for the catalytic action of CcO (3, 22, 24, 49,

53–55) based on spectrophotometric analyses, but none of them were tested against the accompanied enzyme kinetic data. Using a mathematical modeling approach, we have tested various alternative kinetic schemes presented in the literature against relevant experimental data on the CcO

function derived from isolated mitochondrial studies. Although isolated mitochondrial experiments have their own drawbacks (50, 51), they are better suited to identify the kinetics of CcO in its native form compared with the kinetics studies obtained from the purified enzyme in detergent solutions (18).

We have developed a hybrid catalytic scheme based on the scheme proposed by Wilson and coworkers (50–55) by incorporating the details for accurate H^+ balance and pumping. We assumed each electron transfer step in the catalytic cycle is accompanied by H^+ pumping, thereby modifying the associated rate constant for that reaction (Fig. 1) as shown in equation 5. The studies of Wilson and coworkers used a lumped energy state (Q) to describe isolated mitochondrial conditions instead of PMF. We have identified the correlation between Q and PMF (described in METHODS). This way the model has the advantage of a broader applicability for integration into existing mitochondrial bioenergetic models. Furthermore and most importantly, under in vivo conditions, the CcO activity is modified by the presence of NO under both normal and pathological conditions (10, 21, 32). The developed hybrid model is able to describe the inhibitory action of NO on the CcO kinetics, thereby further improving the model utility under different experimental conditions.

The model fittings to the available kinetic data in the present study (Figs. 2 and 3) were considerably improved compared with the previous models (32, 55) because of the accurate H^+ stoichiometry and pumping mechanism. Interestingly, due to such accurate depiction of the CcO kinetics in the absence of NO, addition of only two simple rapid equilibrium interactions of NO, both at the oxidized (E_{1p}) and reduced (E_{2p}) binuclear center, enabled the model to accurately depict the experimentally observed NO inhibition of the CcO function (14, 32) (Fig. 3). The estimated NO dissociation constants for the reduced (K_{il}) and oxidized (K_{i2}) binuclear centers are 40 and 7 nM, respectively, which are comparable to the reported apparent NO inhibition constants (32).

Our model simulations further revealed crucial insights into both the efficiency and control mechanisms involved in the CcO catalytic action. Under completely uncoupled conditions (like purified enzyme), CcO shows very high O_2 consumption rate (~ 170 Torr/s) at high f_{red} , which was drastically decreased upon increased PMF, indicating the effect of proton back pressure exerted on the CcO catalytic action (Figs. 2 and 4). Therefore, accounting for the accurate proton balance and pumping mechanism is crucial for the identification of the role of CcO in various pathophysiological conditions, e.g., ischemia-reperfusion injury, where all of the variables change appreciably with O_2 unavailability. Furthermore, it was also shown in neuronal cultures that NO from neuronal nitric oxide synthase (nNOS) sensitized to hypoxia-induced neuronal death via NO inhibition of CcO (23). The model predictions (Fig. 4C) clearly show that, under in vivo physiological conditions, the O_2 consumption rate significantly drops compared with the unphysiologically high f_{red} and low PMF conditions (Fig. 4, A and B). These results indicate the immense capacity of the CcO to accelerate the O_2 consumption rate under depolarized conditions, an inherent property of the system that was also observed under isolated mitochondrial conditions with addition of uncouplers (e.g., FCCP) that maximally depolarize the mitochondria.

Considerable ambiguity exists regarding the apparent K_m of O_2 for CcO under varying physiological and pathophysiological conditions. Based on the identified second-order rate constant for binding of O_2 to the reduced binuclear center ($6 \times 10^8 M^{-1} \cdot s^{-1}$) and a relatively small dissociation rate constant ($10 s^{-1}$), one can argue for a very high affinity for O_2 binding to CcO at below 1 μM equilibrium dissociation constant. In fact, purified enzymatic studies have shown the values of K_m of O_2 for CcO below 1 μM (7, 41, 45). However, these reported values are too low for mitochondria under physiological conditions, and hence cannot respond to small variations in $[O_2]$ that occur in the cell. Indeed, it has been observed experimentally by different groups that the apparent K_m values are much higher in various cell types compared with the purified enzyme studies, and they are tissue specific (40, 46). Our model predictions also indicate that the apparent K_m for O_2 is near 1 μM under purified enzymatic conditions (0 PMF and high f_{red}) but significantly increased under well-coupled conditions ($> 10 \mu M$) and are extremely high at unphysiologically high PMF and low f_{red} conditions (Fig. 5B). The apparent K_m value also increased with increasing pH, but the effect depends on the PMF and f_{red} . Furthermore, in the presence of NO (Fig. 5D), the apparent K_m values were further increased with increasing [NO], indicating a significant inhibitory role for NO on the CcO activity under normal and pathophysiological conditions. Thus, it is clear from the model analyses that the apparent K_m of O_2 is a function of mitochondrial bioenergetics variables and was altered under increased energy demands and environmental stress.

Under normal physiological conditions, it was shown that endogenous NO can produce up to $\sim 50\%$ inhibition of cellular respiration by binding to CcO both in competitive and non-competitive modes (32). This leads to reduction in O_2 consumption and indirectly activates mitochondrial signaling by altering the mitochondrial redox state and reactive oxygen species (ROS) production (13). These studies show the importance of NO inhibition on the CcO function and need to incorporate its effect on CcO in the modeling formulation to better understand its role in physiological and pathophysiological conditions. For example, it was believed that hypoxia, NO, ROS, Ca^{2+} , and fatty acids mediate the mitochondrial damage during ischemia-reperfusion injury (9). Thus, incorporation of an updated CcO model with NO inhibition into mitochondrial bioenergetics models (4) can provide more insights into the mitochondrial damage during ischemia-reperfusion injury.

The developed CcO kinetic model further proves to be a valuable tool to understand the effect of different mitochondrial bioenergetic variables on the intermediate CcO states in the O_2 reduction. The steady-state model predictions show that, under highly coupled mitochondrial conditions (i.e., high PMF) and in the presence of saturating PO_2 , the enzyme resides mostly in the intermediate state, E_5 , thereby terminating the enzyme-mediated catalytic cycle leading to negligible cytochrome *c* TN. However, as PMF decreases, the enzyme starts to convert to the initial oxidized state E_1 and to the intermediate O_2 -bound form E_4 , which yields maximum turnover under completely uncoupled conditions (i.e., low PMF). In the case of alkaline pH conditions, the rapid equilibrium conversion of the intermediate state E_5 to E_1 was diminished, leading to decreased E_1 and increased E_3 and E_5 (Fig. 6, B and C), thereby lowering the cytochrome *c* TN. This behavior was

further affected by the presence of NO, which directly alters the availability of the fraction of initial oxidized binuclear center (E_1) for turnover and the fraction of reduced binuclear center (E_2) available for O_2 binding (Fig. 6, *D–F*). Thus, our model analyses show the importance of pH and NO binding on the overall kinetics of the CcO reaction in reducing O_2 .

In conclusion, we have developed a thermodynamically constrained mechanistic mathematical model for the kinetics of CcO that includes accurate details for the proton balance and pumping. The model describes the available CcO kinetic data very well, for both isolated mitochondria and purified enzyme conditions in the presence and absence of NO with a minimum number of unknown parameters. The model realistically and explicitly accounts for the thermodynamic dependence of the electron transport and proton translocation mechanisms through its rate constants. The model predictions show that the CcO reaction depends on the mitochondrial variables of f_{red} , PMF, and pH, which explains the observed variations in the apparent K_m of O_2 for CcO under varying physiological conditions compared with the purified enzyme conditions. The intermediate enzyme states depend on pH and f_{red} levels and play a central role in coupling mitochondrial respiration to the energy state (PMF). The enzyme resides mainly in the O_2 -bound form under uncoupled conditions, which is altered as mitochondria become more polarized, leading to the accumulation of other intermediate enzyme states. Furthermore, due to the accurate depictions of the CcO kinetics using the experimental data from isolated mitochondrial conditions, the developed model was able to accurately describe the inhibition kinetics of NO with a minimal number of NO-mediated intermediate states on the CcO kinetics under diverse experimental conditions. Therefore, we posit that the present CcO model, which incorporates various aspects of CcO function, is easy to incorporate into other existing mitochondrial bioenergetics models to replace the existing CcO models in those models. This can be extremely useful to better mechanistically understand the interplay between the CcO enzyme and mitochondrial function in health and disease.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

V.R.P. and R.K.D. conception and design of research; V.R.P. and R.K.D. performed experiments; V.R.P. and R.K.D. analyzed data; V.R.P., A.K.C., and R.K.D. interpreted results of experiments; V.R.P. prepared figures; V.R.P. drafted manuscript; V.R.P., A.K.C., and R.K.D. edited and revised manuscript; V.R.P., A.K.C., and R.K.D. approved final version of manuscript.

REFERENCES

- Antunes F, Boveris A, Cadenas E. On the mechanism and biology of cytochrome oxidase inhibition by nitric oxide. *Proc Natl Acad Sci USA* 101: 16774–16779, 2004.
- Arnold S, Kadenbach B. The intramitochondrial ATP/ADP-ratio controls cytochrome c oxidase activity allosterically. *FEBS Lett* 443: 105–108, 1999.
- Babcock GT, Wikström M. Oxygen activation and the conservation of energy in cell respiration. *Nature* 356: 301–309, 1992.
- Bazil Jason N, Beard Daniel A, Vinnakota Kalyan C. Catalytic coupling of oxidative phosphorylation, ATP demand, and reactive oxygen species generation. *Biophys J* 110: 962–971, 2016.
- Bazil JN, Pannala VR, Dash RK, Beard DA. Determining the origins of superoxide and hydrogen peroxide in the mammalian NADH:ubiquinone oxidoreductase. *Free Rad Biol Med* 77: 121–129, 2014.
- Belevich I, Bloch DA, Belevich N, Wikström M, Verkhovsky MI. Exploring the proton pump mechanism of cytochrome c oxidase in real time. *Proc Natl Acad Sci USA* 104: 2685–2690, 2007.
- Bienfait HF, Jacobs JMC, Slater EC. Mitochondrial oxygen affinity as a function of redox and phosphate potentials. *Bioenergetics* 376: 446–457, 1975.
- Blackmore RS, Greenwood C, Gibson QH. Studies of the primary oxygen intermediate in the reaction of fully reduced cytochrome oxidase. *J Biol Chem* 266: 19245–19249, 1991.
- Borutaite V, Toleikis A, Brown GC. In the eye of the storm: mitochondrial damage during heart and brain ischaemia. *FEBS J* 280: 4999–5014, 2013.
- Brown GC, Borutaite V. Nitric oxide and mitochondrial respiration in the heart. *Cardiovasc Res* 75: 283–290, 2007.
- Chance B, Saronio C, Leigh Jr JS. Functional intermediates in the reaction of membrane bound cytochrome oxidase with oxygen. *J Biol Chem* 250: 9226–9237, 1975.
- Cleeter MWJ, Cooper JM, Darley-Usmar VM, Moncada S, Schapira AHV. Reversible inhibition of cytochrome c oxidase, the terminal enzyme of the mitochondrial respiratory chain, by nitric oxide. *FEBS Lett* 345: 50–54, 1994.
- Cooper CE, Giulivi C. Nitric oxide regulation of mitochondrial oxygen consumption II: molecular mechanism and tissue physiology. *Am J Physiol Cell Physiol* 292: C1993–C2003, 2007.
- Cooper CE, Mason MG, Nicholls P. A dynamic model of nitric oxide inhibition of mitochondrial cytochrome c oxidase. *Biochem Biophys Acta* 1777: 867–876, 2008.
- Cortassa S, Aon MA, Winslow RL, O'Rourke B. A mitochondrial oscillator dependent on reactive oxygen species. *Biophys J* 87: 2060–2073, 2004.
- Gibson QH, Greenwood C. Reactions of cytochrome oxidase with oxygen and carbon monoxide. *Biochem J* 86: 541–554, 1963.
- Golub AS, Pittman RN. Oxygen dependence of respiration in rat spinotrapezius muscle in situ. *Am J Physiol Heart Circ Physiol* 303: H47–H56, 2012.
- Gorbikova EA, Wikström M, Verkhovsky MI. The protonation state of the cross-linked tyrosine during the catalytic cycle of cytochrome c oxidase. *J Biol Chem* 283: 34907–34912, 2008.
- Greenwood C, Gibson QH. The reaction of reduced cytochrome c oxidase with oxygen. *J Biol Chem* 242: 1782–1787, 1967.
- Hagen T, Taylor CT, Lam F, Moncada S. Redistribution of intracellular oxygen in hypoxia by nitric oxide: Effect on HIF1 α . *Science* 302: 1975–1978, 2003.
- Heales SJR, Bolaños JP, Stewart VC, Brookes PS, Land JM, Clark JB. Nitric oxide, mitochondria and neurological disease. *Biochem Biophys Acta* 1410: 215–228, 1999.
- Ishigami I, Hikita M, Egawa T, Yeh SR, Rousseau DL. Proton translocation in cytochrome c oxidase: insights from proton exchange kinetics and vibrational spectroscopy. *Biochem Biophys Acta* 1847: 98–108, 2015.
- Jekabsone A, Neher JJ, Borutaite V, Brown GC. Nitric oxide from neuronal nitric oxide synthase sensitises neurons to hypoxia-induced death via competitive inhibition of cytochrome oxidase. *J Neurochem* 103: 346–356, 2007.
- Konstantinov AA. Cytochrome c oxidase: Intermediates of the catalytic cycle and their energy-coupled interconversion. *FEBS Lett* 586: 630–639, 2012.
- Korzeniewski B. Simulation of oxidative phosphorylation in hepatocytes. *Biophys Chem* 58: 215–224, 1996.
- Krab K, Kempe H, Wikström M. Explaining the enigmatic KM for oxygen in cytochrome c oxidase: A kinetic model. *Biochem Biophys Acta* 1807: 348–358, 2011.

27. Lindsay JG, Owen CS, Wilson DF. The invisible copper of cytochrome c oxidase. pH and ATP dependence of its midpoint potential and its role in the oxygen reaction. *Arch Biochem Biophys* 169: 492–505, 1975.
28. Lindsay JG, Wilson DF. Reaction of cytochrome c oxidase with CO: involvement of the invisible copper. *FEBS Lett* 48: 45–49, 1974.
29. Mackey LN, Kuwana T, Hartzell CR. Evaluation of the energetics of cytochrome c oxidase in the absence of cytochrome C. *FEBS Lett* 36: 326–329, 1973.
30. Markevich NI, Hoek JB. Computational modeling analysis of mitochondrial superoxide production under varying substrate conditions and upon inhibition of different segments of the electron transport chain. *Biochem Biophys Acta* 1847: 656–679, 2015.
31. Mason MG, Nicholls P, Cooper CE. The steady-state mechanism of cytochrome c oxidase: redox interactions between metal centres. *Biochem J* 422: 237–246, 2009.
32. Mason MG, Nicholls P, Wilson MT, Cooper CE. Nitric oxide inhibition of respiration involves both competitive (heme) and noncompetitive (copper) binding to cytochrome c oxidase. *Proc Natl Acad Sci USA* 103: 708–713, 2006.
33. Moser CC, Farid TA, Chobot SE, Dutton PL. Electron tunneling chains of mitochondria. *Biochem Biophys Acta* 1757: 1096–1109, 2006.
34. Murphy MP, Brand MD. Variable stoichiometry of proton pumping by the mitochondrial respiratory chain. *Nature* 329: 170–172, 1987.
35. Pannala VR, Bazil JN, Camara AKS, Dash RK. A biophysically based mathematical model for the catalytic mechanism of glutathione reductase. *Free Rad Biol Med* 65: 1385–1397, 2013.
36. Pannala VR, Bazil JN, Camara AKS, Dash RK. A mechanistic mathematical model for the catalytic action of glutathione peroxidase. *Free Rad Res* 48: 487–502, 2014.
37. Pannala VR, Beard DA, Dash RK. Letter to the Editor: Mitochondrial cytochrome c oxidase: Mechanism of action and role in regulating oxidative phosphorylation. *J Appl Physiol* 119: 157, 2015.
38. Pannala VR, Dash RK. Mechanistic characterization of the thioredoxin system in the removal of hydrogen peroxide. *Free Rad Biol Med* 78: 42–55, 2015.
39. Papa S, Capitanio N, Capitanio G, Palese LL. Protonmotive cooperativity in cytochrome c oxidase. *Biochem Biophys Acta* 1658: 95–105, 2004.
40. Patel SP, Katyare SS. Differences in kinetic properties of cytochrome oxidase in mitochondria from rat tissues. A comparative study. *J Biosci* 60: 785–791, 2005.
41. Petersen LC, Nicholls P, Degn H. The effect of energization on the apparent Michaelis-Menten constant for oxygen in mitochondrial respiration. *Biochem J* 142: 247–252, 1974.
42. Qi F, Dash RK, Han Y, Beard DA. Generating rate equations for complex enzyme systems by a computer-assisted systematic method. *Bioinformatics* 10: 1–9, 2009.
43. Shiva S, Oh JY, Landar AL, Ulasova E, Venkatraman A, Bailey SM, Darley-Usmar VM. Nitroxia: The pathological consequence of dysfunction in the nitric oxide-cytochrome c oxidase signaling pathway. *Free Rad Biol Med* 38: 297–306, 2005.
44. Siletsky SA, Pawate AS, Weiss K, Gennis RB, Konstantinov AA. Transmembrane charge separation during the ferryl-oxo $\rightarrow$ oxidized transition in a nonpumping mutant of cytochrome c oxidase. *J Biol Chem* 279: 52558–52565, 2004.
45. Sinjorgo KMC, Steinebach OM, Dekker HL, Muijsers AO. The effects of pH and ionic strength on cytochrome c oxidase steady-state kinetics reveal a catalytic and a non-catalytic interaction domain for cytochrome c. *Bioenergetics* 850: 108–115, 1986.
46. Tamura M, Hazeki O, Nioka S, Chance B. In vivo study of tissue oxygen metabolism using optical and nuclear magnetic resonance spectroscopies. *Ann Rev Physiol* 51: 813–834, 1989.
47. Tewari SG, Camara AKS, Stowe DF, Dash RK. Computational analysis of Ca^{2+} dynamics in isolated cardiac mitochondria predicts two distinct modes of Ca^{2+} uptake. *J Physiol* 592: 1917–1930, 2014.
48. Tsukihara T, Aoyama H, Yamashita E, Tomizaki T, Yamaguchi H, Shinzawa-Itoh K, Nakashima R, Yaono R, Yoshikawa S. The whole structure of the 13-subunit oxidized cytochrome c oxidase at 2.8 Å. *Science* 272: 1136–1144, 1996.
49. Wikström M, Verkhovsky MI. Mechanism and energetics of proton translocation by the respiratory heme-copper oxidases. *Biochem Biophys Acta* 1767: 1200–1214, 2007.
50. Wilson DF, Harrison DK, Vinogradov A. Mitochondrial cytochrome c oxidase and control of energy metabolism: Measurements in suspensions of isolated mitochondria. *J Appl Physiol* 117: 1424–1430, 2014.
51. Wilson DF, Harrison DK, Vinogradov SA. Oxygen, pH, and mitochondrial oxidative phosphorylation. *J Appl Physiol* 113: 1838–1845, 2012.
52. Wilson DF, Lee WMF, Makonnen S, Finikova O, Apreleva S, Vinogradov SA. Oxygen pressures in the interstitial space and their relationship to those in the blood plasma in resting skeletal muscle. *J Appl Physiol* 101: 1648–1656, 2006.
53. Wilson DF, Owen CS, Erecińska M. Quantitative dependence of mitochondrial oxidative phosphorylation on oxygen concentration: a mathematical model. *Arch Biochem Biophys* 195: 494–504, 1979.
54. Wilson DF, Owen CS, Holian A. Control of mitochondrial respiration: A quantitative evaluation of the roles of cytochrome c and oxygen. *Arch Biochem Biophys* 182: 749–762, 1977.
55. Wilson DF, Vinogradov SA. Mitochondrial cytochrome c oxidase: Mechanism of action and role in regulating oxidative phosphorylation. *J Appl Physiol* 117: 1431–1439, 2014.
56. Woodbury JW. Eyring rate theory model of the current-voltage relationships of ion channels in excitable membranes. In: *Advances in Chemical Physics*. New York, NY: Wiley, 2007, p. 601–617.
57. Wu F, Yang F, Vinnakota KC, Beard DA. Computer modeling of mitochondrial tricarboxylic acid cycle, oxidative phosphorylation, metabolite transport, and electrophysiology. *J Biol Chem* 282: 24525–24537, 2007.